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ISOMER YIELD RATIOS OF PHOTONUCLEAR REACTIONS AT $E_{\gamma max}$ 25 AND 30 MeV

S. R. Palvanov and O. Razhabov

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The study of isomer ratios, i.e., the relative probabilities of occupying isomer m and ground g levels of the final nucleus in photonuclear reactions, makes it possible to determine the character of the spin dependence of the density of nuclear levels and to understand the reaction mechanism better. In addition, data on the isomer yield ratios Y_m/Y_g are necessary for γ activation analysis, which is one of the applied directions of nuclear physics. According to the data of [1], the reactions leading to the formation of isomer states of the final nuclei comprise ~40% of all cases which are of interest for this method of analysis. Without taking account of the data on the isomer yield ratios Y_m/Y_g of photonuclear reactors, it is impossible to study the analytic possibilities of the method and to develop optimal methods of analysis of the composition of matter.

In the present work the isomer yield ratios Y_m/Y_g of photonuclear reactions of the type (γ, n), (γ, p), and $(\gamma, 2n)$ on nuclei with $45 \le A \le 197$ with maximum bremsstrahlung energy $E_\gamma$max of 25 and 30 MeV are determined.

The choice of 25 and 30 MeV as the maximum bremsstrahlung energies is not accidental. It is made because most experimental works studying the yields of photonuclear reactions for γ-activation analysis were performed at this energy. In addition, for $E_\gamma \ge E_m + \Gamma$ (E_m, and Γ are, respectively, the position of the maximum and the half-width of the giant dipole resonance) the condition $\sigma_m/\sigma_g \equiv Y_m/Y_g = \text{const}$ [2] holds for most photonuclear reactions (σ_m/σ_g are the isomeric ratios of the reaction cross sections), i.e., in the energy range beyond the giant dipole resonance, the isomer ratios of the cross sections and yields do not change sharply.

The measurements were performed in the bremsstrahlung beam of the SB-50 high-current betatron [3] by the method of induced activity. To increase the bremsstrahlung flux density, samples in the form of standard pellets, 17 mm in diameter, consisting of chemically pure substances in their elementary form, as well as in the form of compounds, were irradiated inside the accelerator chamber in a special apparatus, placed 12 cm from the tungsten bremsstrahlung target and placed together with the K5-2A pneumatic transport setup (the delivery time of the samples to irradiation ~4 sec) [4]. The apparatus makes it possible to irradiate samples up to 50 g in mass. To take account of the effect of changes in the beam intensities, a monitor – copper foil – was irradiated together with the sample.

The temporal regimes, i.e., the irradiation time, the pauses, and the measurement time, were chosen in accordance with the half life of the radionuclides formed. The induced γ activity was measured with a DGDK-63V Ge(Li) detector with measuring-channel resolution ~3.5 keV for the 1332 keV line of ^{60}Co. Spectrometric information was recorded and processed using an LP-4900 (NOKIA) 4096-channel pulse-height analyzer. The energy calibration of the γ spectrometer and the determination of its efficiency as a function of the γ-ray energy were performed using a collection of OSGI sources. The efficiency of the γ spectrometer was determined by the method described in [5].

The population of the isomer m and ground state g levels of the daughter nuclei was identified according to the corresponding γ lines. The spectroscopic characteristics of the levels of the nuclei studied were taken from [6].

The isomer yield ratios of the photonuclear reactions were determined by comparing the areas under the photopeaks of the corresponding γ lines [7]. For the reactions ^{59}Co$(\gamma, n)^{58m,g}$Co and ^{81}Br$(\gamma, n)^{80m,g}$Br, the values of Y_m/Y_g were determined by decomposing the decay-accumulation curve, measured according to one γ line 810 and 817 keV, respectively. The experimental data for these reactions were analyzed by the least-squares method using an ES1841 computer.

The results obtained and their measurement error are presented in Table 1. The measurement errors for the isomeric yield ratios were estimated on the basis of the statistics of the counts in the photopeak of the γ line measured. Data on Y_m/Y_g obtained by other authors in the energy range 20–30 MeV are presented in the table for comparison. Here, in the case of the

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TABLE 1. Isomer Yield Ratios of Photonuclear Reactions

Reaction	J^{π}; $T_{1/2}$		$E_{\gamma max}$, MeV	Y_m/Y_g	Source
	m	g			
1	2	3	4	5	6
$^{45}Sc(\gamma, n)^{44}Sc$	6^+; 2.44 days	2^+; 3.93 h	22	0.23 ± 0.02	[1]
			25	0.18 ± 0.04	This work
			30	0.24 ± 0.03	»
$^{47}Ti(\gamma, p)^{46}Sc$	7^+; 20 sec	4^+; 84.1 days	25	0.16 ± 0.02	»
			30	0.20 ± 0.02	»
$^{59}Co(\gamma, n)^{58}Co$	5^+; 9.2 h	2^+; 70.8 days	25	1.19 ± 0.01	[2]
			25	1.22 ± 0.02	This work
			30	1.30 ± 0.02	»
$^{76}Ge(\gamma, n)^{75}Ge$	$7/2^+$; 48 sec	$1/2^-$; 82.8 min	22	1.25 ± 0.05	[8]
			25	1.26 ± 0.07	This work
			30	1.27 ± 0.05	»
$^{82}Se(\gamma, n)^{81}Se$	$7/2^+$; 57.3 min	$1/2^-$; 18.5 min	22	0.54 ± 0.03	[2]
			25	0.57 ± 0.03	This work
			30	0.56 ± 0.02	»
$^{81}Br(\gamma, n)^{80}Br$	5^-; 4.42 h	1^+; 17.6 min	30	0.48 ± 0.03	[1]
			25	0.46 ± 0.03	This work
			30	0.48 ± 0.02	»
$^{85}Rb(\gamma, n)^{84}Rb$	6^+; 20.5 min	2^-; 32.9 days	22	0.31 ± 0.02	[1]
			25	0.33 ± 0.01	This work
			30	0.35 ± 0.01	»
$^{86}Sr(\gamma, n)^{85}Sr$	$1/2^-$; 68 min	$9/2^+$; 64.8 days	22	0.14 ± 0.02	[1]
			30	1.78 ± 0.25	[2]
			25	0.64 ± 0.02	This work
			30	0.54 ± 0.03	»
$^{89}Y(\gamma, 2n)^{87}Y$	$9/2^+$; 13 h	$1/2^-$; 80.3 h	23	0.25 ± 0.03	[9]
			25.6	0.35 ± 0.04	[9]
			28.6	0.43 ± 0.04	[9]
			25	0.33 ± 0.03	This work
			30	0.45 ± 0.04	»
$^{90}Zr(\gamma, n)^{89}Zr$	$1/2^-$; 4.18 min	$9/2^+$; 78.4 h	22	1.44 ± 0.02	[1]
			25	1.21 ± 0.04	This work
			30	1.35 ± 0.05	»
$^{96}Mo(\gamma, p)^{95}Nb$	$1/2^-$; 87 h	$9/2^+$; 35 days	25	0.55 ± 0.05	»
			30	0.56 ± 0.06	»
$^{98}Mo(\gamma, n)^{97}Nb$	$1/2^-$; 1 min	$9/2^+$; 72.1 min	22	0.71 ± 0.02	[1]
			20–25	0.58 ± 0.11	[2]
			25	0.52 ± 0.02	This work
			30	0.53 ± 0.02	»
$^{113}In(\gamma, n)^{112}In$	4^+; 20.9 min	1^+; 14.4 min	25	4.73 ± 0.08	»
			30	4.12 ± 0.05	»
$^{113}In(\gamma, 2n)^{111}In$	$1/2^-$; 7.6 min	$9/2^+$; 2.83 days	25	0.14 ± 0.03	»
			30	0.13 ± 0.01	»

Table 1 (continued)

1	2	3	4	5	6
$^{112}Sn(\gamma, p)^{111}In$	$1/2^-$; 7.6 min	$9/2^+$; 2.83 days	22	0.044 ± 0.001	[1]
			25	0.043 ± 0.002	This work
			30	0.047 ± 0.003	»
$^{118}Sn(\gamma, p)^{117}In$	$1/2^-$; 1.94 h	$9/2^+$; 44 min	22	2.63 ± 0.18	[10]
			25	2.9 ± 0.2	This work
			30	3.1 ± 0.2	»
$^{120}Te(\gamma, n)^{119}Te$	$11/2^-$; 4.68 days	$1/2^+$; 16.05 h	25	1.04 ± 0.02	»
			30	1.06 ± 0.02	»
$^{130}Te(\gamma, n)^{129}Te$	$11/2^-$; 33.5 days	$3/2^+$; 69 min	20	0.45 ± 0.06	[11]
			25	0.49 ± 0.03	This work
			30	0.48 ± 0.03	»
$^{138}Ce(\gamma, n)^{137}Ce$	$11/2^-$; 34.4 h	$3/2^+$; 9 h	25	0.12 ± 0.02	»
			30	0.13 ± 0.02	»
$^{140}Ce(\gamma, n)^{139}Ce$	$11/2^-$; 56 sec	$3/2^+$; 139.38 days	23.8	0.182 ± 0.018	[12]
			31	0.193 ± 0.014	[12]
			25	0.190 ± 0.017	This work
			30	0.201 ± 0.015	»
$^{142}Nd(\gamma, n)^{141}Nd$	$11/2^-$; 61.3 sec	$3/2^+$; 2.42 h	18	0.046 ± 0.003	[13]
			25	0.043 ± 0.002	This work
			30	0.045 ± 0.001	»
$^{197}Au(\gamma, n)^{196}Au$	12^-; 9.7 h	2^-; 6.18 days	52	$(6.1 \pm 0.4)10^{-4}$	[14]
			25	$(5.4 \pm 0.3)10^{-4}$	This work
			30	$(5.5 \pm 0.4)10^{-4}$	»

reaction (γ, n) on ^{197}Au, the values of Y_m/Y_g are presented for $E_{\gamma max}$ 52 MeV. The experimental data on the isomer yield ratios Y_m/Y_g of photonuclear reactions of the type (γ, n) on the nuclei ^{45}Sc, ^{59}Co, ^{76}Ge, ^{82}Se, ^{81}Br, 85Rb, ^{140}Ce, ^{197}Au, and $(\gamma, 2n)$ on the ^{89}Y nucleus and (γ, p) on the nuclei ^{98}Mo and ^{112}Sn agree, to within the measurement error, with the results of [2], which include the experimental data obtained by different authors up to 1987. For the (γ, n) reaction on ^{86}Sr, the data obtained by different authors do not agree with one another.

For the photonuclear reactions $^{47}Ti(\gamma, p)^{46}Sc$, $^{87}Rb(\gamma, n)^{86}Rb$, $^{96}Mo(\gamma, p)^{95}Nb$, $^{113}In(\gamma, n)^{112}In$, $^{113}In(\gamma, 2n)^{111}In$, $^{120}Te(\gamma, n)^{119}Te$, and $^{138}Ce(\gamma, n)^{137}Ce$, the values of Y_m/Y_g have been obtained for the first time.

The values of Y_m/Y_g for the photonuclear reactions $^{118}Sn(\gamma, p)^{117}In$, $^{130}Te(\gamma, n)^{129}Te$, and $^{142}Nd(\gamma, n)^{141}Nd$ agree, to within the measurement error, with the results of [10, 11, 13], respectively.

The experimental results obtained in this work on isomer yield ratios of photonuclear reactions can be used to assess the fundamental possibilities of the method of γ-activation analysis on a specific analytical system and for development of methods of analysis and for planning experiments studying the isomer ratios in photonuclear reactions.

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